

BINAY – Lighting Up The Future

Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 43m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



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Self-navigating robots

Robots could navigate through constantly changing surroundings with virtually no input from humans, thanks to a system being developed by researchers at MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL).



The researchers used a PR2 robot, developed by Willow Garage, with Microsoft's Kinect sensor to test their system

The approach is based on simultaneous localisation and mapping (SLAM) technique. As the robot travels through an unexplored area, the Kinect sensor's visible-light video camera and infrared depth sensor scan the surroundings, building up a 3D model of the walls of the room and the

objects within it. Then, when the robot passes through the same area again, the system compares the features of the new image it has created — including details such as the edges of walls, for example — with all the previous images it has taken until it finds a match.

At the same time, the system constantly estimates the robot's motion, using on-board sensors that measure the distance its wheels have rotated. By combining the visual information with this motion data, it can determine where within the building the robot is positioned. Combining the two sources of information allows the system to eliminate errors that might creep in if it relied on the robot's on-board sensors alone.

Low-cost solar cell created from plants

Researcher at the Massachusetts Institute of Technology have turned agricultural waste such as cut grass and dead leaves into low-tech, inexpensive solar panels.

In earlier research, a complex of molecules known as photosystem-I (PS-I)—the tiny structures within plant cells that carry out photosynthesis—was derived. It was stabilised chemically to form a layer on a glass substrate that could produce an electric current when exposed to light (like a conventional photovoltaic cell).

In the new research, the efficiency was improved by exposing much more of the PS-I complex per surface area of the device to the sun. Researchers created on a chip a microscopic forest of zinc-oxide (ZnO) nanowires as well as a sponge-like titanium dioxide (TiO₂) nanostructure coated with the light-collecting material derived from bacteria, with PS-I coating the 'trees' from top to bottom. The nanowires served as a supporting structure for the material, and as wires to carry the flow of electrons generated by the molecules down to the supporting layer of material, from which it could be connected to a circuit creating electric current.